Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2019-P2-Model)** **Date: 15-04-2026**

Time: 02.30Pm to 05.30Pm **GTA-24** **Max. Marks: 186**

KEY SHEET

PHYSICS

1	**BC**	2	**ABC**	3	**AD**	4	**ABC**	5	**AD**	6	**BC**
7	**ABC**	8	**ABCD**	9	**9.08-9.13**	10	**90-92**	11	**2**	12	**9**
13	**1.48-1.53**	14	**0.98-1.02**	15	**B**	16	**B**	17	**C**	18	**D**

CHEMISTRY

19	**BCD**	20	**BCD**	21	**ABD**	22	**ABCD**	23	**ACD**	24	**BC**
25	**ABCD**	26	**BCD**	27	**24**	28	**6**	29	**100**	30	**530**
31	**6 TO 14**	32	**7**	33	**D**	34	**C**	35	**C**	36	**C**

MATHS

37	**ABCD**	38	**C**	39	**ABD**	40	**BC**	41	**ACD**	42	**ABCD**
43	**AC**	44	**AB**	45	**25**	46	**1**	47	**4046**	48	**12**
49	**6**	50	**1**	51	**A**	52	**C**	53	**A**	54	**D**

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SOLUTIONS

PHYSICS

1. $s = -\int_{L}^{0} \frac{1}{r} dl = \frac{L^2}{2r} = 1.6\,m$

Therefore for the time elapsed until the impact. We obtain:

$T = \frac{L^2}{2rv} = 4s$.

The instantaneous radius of curvature is l . Therefore the tension that provides the centripetal acceleration has magnitude mv^2 / l . Let us express l as a function of time. To do this, let the upper limit of integration be l instead of 0 and let the distance covered be $s = vt$:

$s = -\int_{L}^{l} \frac{1}{r} dl = \frac{1}{2r}(L^2 - l^2) = vt$

From which:

$$l = \sqrt{L^2 - 2rvt}$$

The force as a function of time is therefore:

$$F = \frac{mv^2}{\sqrt{L^2 - 2rvt}}$$

2. $\frac{1}{30} = \frac{1}{V} + \frac{1}{90} \Rightarrow V = 45$ (for convex) $m = \frac{-1}{2}$

(For concave) $\frac{-1}{50} = \frac{1}{V} - \frac{1}{25} \Rightarrow V = 50 \Rightarrow m = \pm 2$

$\therefore$ Final image is at x = 160 cm

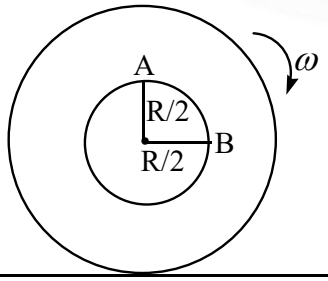
$\therefore$ Real image can be at $(135, -0.75)$

3. $T = 2\pi\sqrt{LC} = 2\pi\sqrt{2\pi\varepsilon_0 R}$

After this time, the charges on the spheres become equal to their initial values. When the charge of the first sphere is $Q/2$, the charge of the second sphere is $Q/2$ as well, so the voltage across the coil is zero, and the time elapsed until this state is a quarter of the period:

$l = \frac{T}{4} = \frac{\pi}{2}\sqrt{2\pi\varepsilon_o LR}$

4.



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$V = RW$ (linear velocity)

$a_{om} = 0$

$\vec{a}_A = \frac{R}{2}W^2 \downarrow$

$\vec{a}_B = \frac{R}{2}W^2(\leftarrow)$

No two points on the disc have same acc.

5. For an exothermic reaction

$X + x \rightarrow Y + y$

If K_i is the kinetic energy of incident particle x, them from energy conservation,

$K_i + (m_x + M_x)c^2 = K_Y + K_y + (M_Y + m_y)c^2$

$K_Y + K_y = K_i + (m_x + M_X - M_Y - m_y)c^2$

$K_Y + K_y = K_i + Q$

In any exothermic reaction, mass of the products is less then the mass of reactants, i.e., in products, the nucleons are more tightly bound and hence have greater BE per nucleon as minimum energy given to the reactant must be greater than Q value.

6. $a = \frac{F}{m} = \frac{eE}{m} = \frac{e.\frac{1}{2r_A\pi}\frac{\alpha\phi}{\Delta t}}{m} = \frac{eR^2\pi\ \Delta B}{m2r_A\pi\ \Delta t} = \frac{e}{m}\frac{R^2\ \Delta B}{2r_A\Delta t}$

Calculating we will get approximately

$9.38\times10^9 \frac{m}{s^2}$.

$v = \sqrt{2as} = \sqrt{2\frac{eE}{m}r\varphi} = \sqrt{\frac{eR^2\Delta B}{m\Delta t}\varphi}$,

Since the magnitude of the electric filed E is constant along the arc and its direction is tangential, so the electron will move along the arc with uniformly changing speed.

Solving we will get approximately

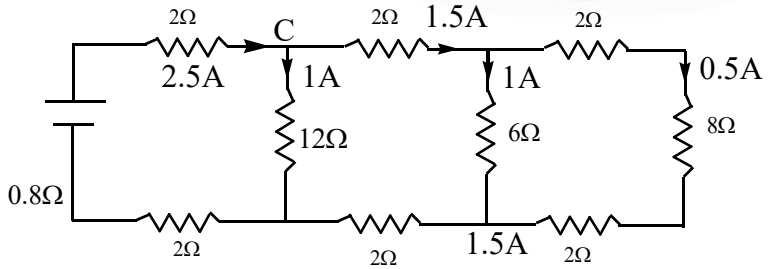
$34.3\frac{km}{s}$

7. $\Delta x = n\lambda$

$5000\times10^{-9} = n\lambda$

$\lambda = \frac{5000}{n}nm$

8.



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9. Say original position of the lower piston from bottom is x. from the first process we can find x = 8.63 cm

With the help of this result, the problem's question can be easily answered. After removing the upper piston from the cylinder, the pressure of the gas below the other piston becomes the sum of the external pressure and the pressure due to the weight of the piston: 110000 Pa. The new volume is determined by Boyle's law:

$$p_1.A.x = p_2^n.A.z$$

The cross section cancels out. With the numerical data the equation reads:

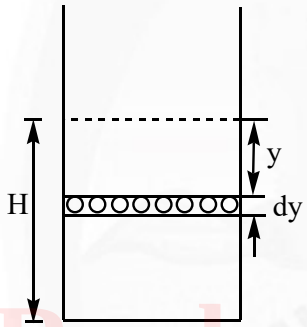
$$120000\,Pa.0.0863m = 110000\,Pa.z\,,$$

and the new height of the piston is:

$$z = \frac{12Pa.0.0863m}{11Pa} = 0.0941m = 9.11cm\,.$$

10. Speed of efflux at a depth y below the water surface is $v = \sqrt{2gy}$

Consider a strip of height dy as shown.



Cumulative area of holes in this strip is $A = \eta(2\pi rdy)$

Rate (volume per unit time) at which the liquid flows out through the strip is

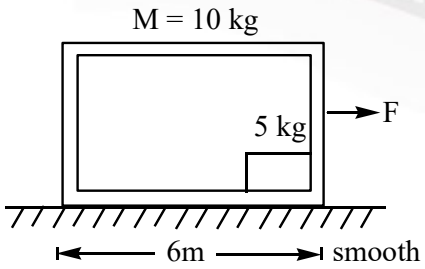
$= vA = 2\pi\eta r\sqrt{2g}\sqrt{ydy}$

Total rate of outflow $= 2\pi\eta r\sqrt{2g}\int_0^H \sqrt{ydy}$

$$= 2\pi\eta r\sqrt{2g}\frac{2}{3}\left[y^{\frac{3}{2}}\right]^H = \frac{4}{3}\pi\eta r\sqrt{2g}H^{3/2} \qquad \therefore \text{ For level to remain static}$$

$$Q = \frac{4}{3}\pi\eta r\sqrt{2g}H^{3/2} = 66.4$$

11.



$$a_{system} = \frac{F}{10+5} = \frac{F}{15}$$

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Pseudo force on 5kg $5\left(\frac{F}{15}\right)=\frac{F}{3}$

$\frac{F}{3}=\mu mg \Rightarrow \frac{F}{3}=0.2\times5\times10$ $\qquad F=10\times3=30N$

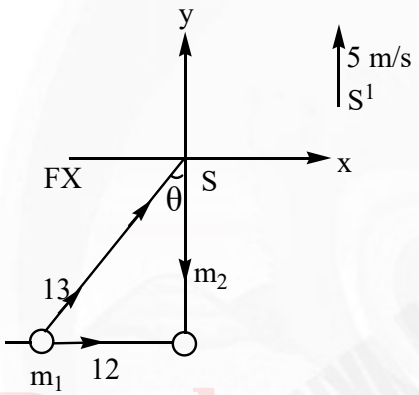
If force applied $=2F=60N$ $\qquad F_{net}=m_1a_1+m_2a_2$

$60=5\times2+10\times a$ (because $a_{\max}$ for 5kg $=2m/s^2$)

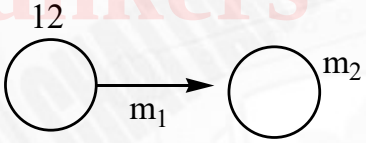
$\frac{60-10}{10}=a$ $\qquad 5=a_{h\ b}$

$a_{rel}=3m/s^2$ $\qquad s_{rel}=\frac{1}{2}a_{rel}t^2$ $\qquad t=\sqrt{\frac{2S_{rel}}{a_{rel}}}=\sqrt{\frac{2\times6}{3}}=2\text{sec}$

12.



(a) In the frame of reference of observer.



after collision



so apply momentum conservation eqn.

$12m_1+0=0+m_2V_2$ ……..(1)

apply (m.e.c)

$\frac{1}{72}\times\frac{1}{2}\times m_1\times(12)^2=\frac{1}{2}\times m_1\times(12)^2-\frac{1}{2}\times m_2\times(V_2)^2$ …….(2)

$\Rightarrow$ put value of $\frac{m_1}{m_2}$ in (Solve get) $\frac{m_1}{m_2}$. $\Rightarrow\frac{1}{72}\times\frac{1}{2}\times\frac{m_1}{m_2}\times(12)^2=\frac{1}{2}\left(\frac{m_1}{m_2}\right)\times12^2-\frac{1}{2}(V_2)^2$

$\Rightarrow\frac{1}{72}\times\frac{1}{2}\times\frac{V_2}{12}\times(12)^2=\frac{1}{2}\frac{V_2}{12}\times12^2-\frac{1}{2}\times2^2$ on solving we get $V_2=9m/\text{sec}$

13. $\frac{r\omega^2}{g}=\frac{R-r}{l-h}$

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The distance r of the bead from the axis of rotation (the radius of its circular orbit) is unknown. This can be determined from the length of the thread and the given sinking of the bead. Since the density of the bead is smaller than the density of water, it will move inwards. Using the Pythagorean theorem

$$r = R - \sqrt{l^2 - (l-h)^2} = R - \sqrt{(2l-h).h}$$

Substituting this into the previous equation and rearranging for ω gives

$$\omega = \sqrt{\frac{g\sqrt{(2l-h).h}}{(l-h).\left(R - \sqrt{(2l-h).h}\right)}} = 9.68\ s^{-1}$$

and the requested number of revolutions is

$$n = \frac{\omega}{2\pi} = 1.51 s^{-1}$$

14. The time difference between the start of the fall and the gun fire is :

$$\Delta t = t - t_1 = \sqrt{\frac{h}{g} + \frac{d^2}{v^2}} - \frac{d}{v}$$

15. Fundamental frequency is maximum when length is minimum i.e L_0

$$Case 1. L = L_0, T = T_0; f_1 = \frac{1}{2L_0}\sqrt{\frac{T_0}{\mu}}$$

$$Case 2. f_2 = \frac{1}{L_0}\sqrt{\frac{T_2}{2\mu}} = \frac{f_0}{\sqrt{2}}$$

$$Case 3. f_3 = \frac{1}{L_0}\sqrt{\frac{T_2}{3\mu}} = \frac{f_0}{\sqrt{3}}$$

$$Case 4. f_4 = \frac{1}{L_0}\sqrt{\frac{T_2}{4\mu}} = \frac{f_0}{2}$$

16. **Case 1.**

$L = L_0, T = T_0 . f = f_0$

$$f_1 = \frac{1}{2L_0}\sqrt{\frac{T_0}{\mu}}$$

Case 2.

$$L = \frac{3L_0}{2}$$

$$f_2 = \frac{3}{2 \times \frac{3L_0}{2}}\sqrt{\frac{T_2}{2\mu}} = f_0 = \frac{1}{\sqrt{2}L_0}\sqrt{\frac{T_2}{\mu}} \Rightarrow T_2 = \frac{T_0}{2}$$

Case 3.

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$$L=\frac{5L_0}{4}$$

$$f_3=\frac{5}{2\times\frac{5L_0}{4}}\sqrt{\frac{T_3}{3\mu}}=f_0$$

$$\Rightarrow f_0=\frac{2}{\sqrt{3}L_0}\sqrt{\frac{T_3}{\mu}}\Rightarrow T_3=\frac{3T_0}{16}$$

Case 4.

$$L=\frac{7L_0}{4}$$

$$\Rightarrow f_4=\frac{14}{2\times\frac{7L_0}{4}}\sqrt{\frac{T_4}{4\mu}}=f_0$$

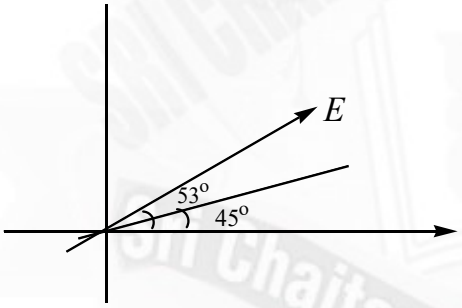
$$\Rightarrow f_0=\frac{2}{L_0}\sqrt{\frac{T_4}{\mu}}\Rightarrow\frac{T_0}{16}$$

17. $I_0=\frac{1}{2}C\in_0 E_0^2=\frac{1}{2}\times3\times10^8\times4\pi\times5^2\in_0$

$$=\frac{3\times25\times10^8}{2\times9\times10^9}=\frac{5}{12}w/m^2$$

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P)



$$I=I_0\cos^2(53-45)=\frac{5}{12}\left[\frac{3}{5}\times\frac{1}{\sqrt{2}}+\frac{4}{5}\times\frac{1}{\sqrt{2}}\right]^2=\frac{5}{12}\times\frac{49}{25}\times\frac{1}{2}=\frac{49}{120}w/m^2$$

(Q) $I=I_0\cos^2 37^0=\frac{4}{15}w/m^2$

(R) $I=I_0\cos^2 53^0=\frac{3}{20}w/m^2$

(S) $I=I_0\cos^2(53-37)=\frac{48}{125}w/m^2$

18. Similar to solution of Q no. 17

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CHEMISTRY

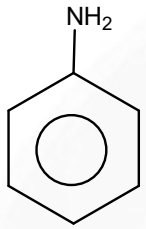
19. A-Ortho nitro phenol is less acidic than para nitro phenol due to intramolecular hydrogen bonding

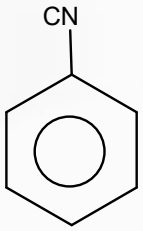
B-ortho effect

C-d-orbital resonance in p-chloro phenol

D-SIR effect

20.

P = 

Q = 

21. Theory based

22. $X_A = \frac{1}{4}, X_B = \frac{3}{4}, P = 500$

$X_A = \frac{1}{6},\ X_B = \frac{5}{6},\ P = 550$

$500 = \frac{1}{4}P_A^0 + \frac{3}{4}P_B^0$

$550 = \frac{1}{6}P_A^0 + \frac{5}{6}P_B^0$

$P_A^0 = 50mm, P_B^0 = 650mm$

Finally when 4 moles of A are also added

$X_A = \frac{1}{2}, X_B = \frac{1}{2}$

$P = \frac{1}{2} \times 50 + \frac{1}{2} \times 650$

$P = 350mm$

23. A) On increasing pressure volume decreases so concentration of each gaseous substance increases

C) K_P remains constant so total pressure at equilibrium remains un affected.

D)Given reaction as an Endothermic reaction so K_P increases with the increase of temparature

24. Covalence of 'N' in $N_2O_5 = 4$

25. $Pb_3O_4 + HCl \longrightarrow PbCl_2 + Cl_2 + H_2O$

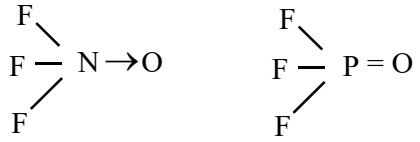
$PbO + HCl \longrightarrow PbCl_2 + H_2O$

$PbO_2 + HCl \longrightarrow PbCl_2 + Cl_2 + H_2O$

26. $(i)\ N_3^- : Same bond\ length$

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(ii) 

$(iii)\ NF_3 < NCl_3\ (\mu value)$

27. 12 representations for D isomer & 12 representations for L isomer (Total = 24)

28. a,b,c,d,e,h

29.

$$Ag^+(aq) + Br^-(aq) \longrightarrow AgBr(s)$$
$$\Delta H = -90KJ$$

$$Ag(s) \xrightarrow{aq} Ag^+(as)$$
$$\Delta H = 5x$$

$$Ag_2(s) + \frac{1}{2}Br_2(l) \longrightarrow AgBr(s)$$
$$\Delta H = -90 + 5x - 6x$$
$$-90 + 5x - 6x = -110$$
$$x = 20$$

Enthalpy of formation of $Ag^+ = 5x$

$= 100\,kJ/mol$

30. $dG = VdP - sdT$

$$\int dG = V\int_{P_1}^{P_2} dP - \int_{T_1}^{T_2} sdT$$

$V\int_{P_1}^{P_2} dP = V[P_2 - P_1]$ At constant volume $\frac{P_1}{T_1} = \frac{P_2}{T_2}$ $\therefore P_2 = \frac{4}{3}bar$

$$V[P_2 - P_1] = 24.6\left(\frac{4}{3} - 1\right) \text{ bar litre}$$

$$= 24.6\left(\frac{1}{3}\right) \times 100\,Joule$$

= 820 Joule

$$\int_{T_1}^{T_2} SdT = \int_{300}^{400} \left(10 + 10^{-2}T\right)dT$$

$$= \int_{300}^{400} 10dT + \int_{300}^{400} 10^{-2}TdT$$

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$$=10\times100+10^{-2}\left[\frac{T^2}{2}\right]_{300}^{400}$$

$$=1000\times10^{-2}\left[\frac{160000}{2}-\frac{90000}{2}\right]$$

$$=1000+10^{-2}\frac{70000}{2}$$

$$=1000+350$$

$$=1350\,Joule$$

$$\Delta G=820-1350$$

$$\Delta G=-530\,Joule$$

$$|\Delta G|=530\,Joule$$

31. $Ag^+ + e^- \rightarrow Ag\ E^0\ (Ag^+/Ag)=0.8$

$Ag(NH_3)_2^+ \rightleftharpoons Ag^+ + 2NH_3, K=6\times10^{-14}$

Add

$Ag(NH_3)_2^+ + e^- \rightarrow Ag(s)+2NH_3$

$\Delta G^0=\Delta G_1^0+\Delta G_2^0$

$-1\times F\times E^0=-1\times F\times0.8-RT\ln K$

On solving

$E^0=+0.01Volt$

$1000\times E^0=10$

32. $x=2,\ z=5$

$CrCl_3, ZnSo_4, \longrightarrow 'x'=2$ (first ppt and then soluble in NaOH)

$yellow: K_3\left[Co(NO_2)_6\right], BaCrO_4, SrCrO_4, PbCrO_4, AgI\ (z=5)$

33 & 34

P-(A = benzoic acid, B = benzyl alcohol)

Q-(A = acetone, B = acetaldehyde)

R-(A = acetophenonel, B = diethylketone)

S-(A = phenol, B = acetone)

35 &36

$Zn \longrightarrow distillation$

$Sn \longrightarrow Liquation$

$Si \longrightarrow$ Zone refining

$Ti \longrightarrow Van\ Arkel\ method$

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MATHS

37. B)

$$\left.\begin{matrix} n(S)=6^3 \\ n(E)=9+9=18 \end{matrix}\right\} P(E)=\frac{18}{6^3}=\frac{3}{36}=\frac{1}{12}$$

(C)

$$\left.\begin{matrix} n(S)=6^4 \\ n(E)=6_{C_2}\times 6_{C_2} \end{matrix}\right\} P(E)=\frac{15\times 15}{6^4}=\frac{25}{144}$$

(D)

$$n(E)=1^2+2^2+3^2+4^2+5^2+6^2+5^2+4^2+3^2+2^2+1^2=2(1+4+9+16+25)+36=146$$

$$P(E)=\frac{146}{6^4}=\frac{73}{648}$$

38. $f(x)$ is continuous where $3\sin x+a^2-10a+30=4\cos x$

Or $a^2-10a+30=4\cos x-3\sin x$

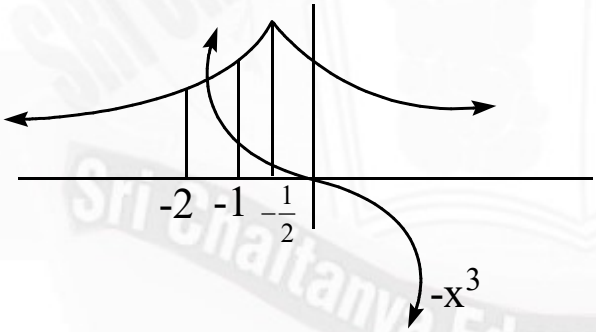
$LHS\geq 5, RHS\leq 5$

$$\Rightarrow (a-5)^2+5=4\cos x-3\sin x$$

$$\Rightarrow a=5, x=2n\pi-\tan^{-1}\left(\frac{3}{4}\right), n\in I$$

39. $f(x)=0\Rightarrow x^3\left(x^2+x+1\right)+\left(x^2+x+1\right)+2=0$

$$\Rightarrow -x^3=1+\frac{2}{x^2+x+1}$$



40. $x^7+ax^5-1=0$ .....(1)

And $x^6+ax^4-4x=0$

$$\Rightarrow x^7+ax^5-4x^2=0 \quad .....(2)$$

Equation $(1)-(2)\Rightarrow 4x^2=1\Rightarrow x=\pm\frac{1}{2}$.

Sum of all possible common root = 0.

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If $x=\frac{1}{2}$ is a solution, then $x=\frac{1}{2}$ satisfy given equation $\left(\frac{1}{2}\right)^7+a\left(\frac{1}{2}\right)^5-1=0$

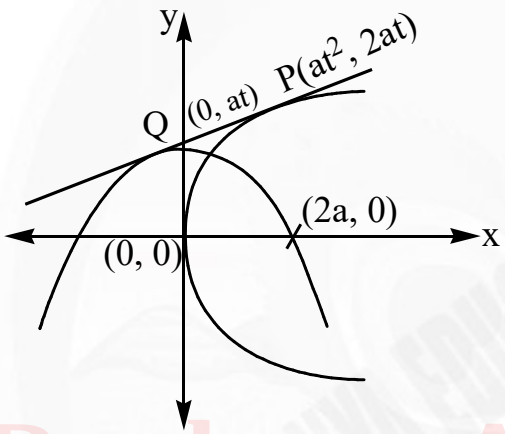
$\Rightarrow 1+4a-128=0\Rightarrow a=\frac{127}{4}$

If $x=\frac{-1}{2}$ is a solution, then $x=\frac{-1}{2}$ satisfy given equation $\left(\frac{-1}{2}\right)^7+a\left(\frac{-1}{2}\right)^5-1=0$

$\Rightarrow a=\frac{-129}{4}$. So, sum of possible value of a is $\frac{-1}{2}$.

41. Tangent at P

$ty=x+at^2$



$Q\equiv(0,at)$

Now, Equation of parabola with vertex (0, at) and (0, 0) as its focus $x^2=-4at(y-at)$

Which passes through (2a, 0)

$4a^2=-4at(-at)\Rightarrow t^2=1\Rightarrow t=\pm 1$

Possible equation curve $C_2: x^2=-4a(y-a)$ or $x^2=4a(y+a)$

Co-ordinate of P are: $(a,\pm 2a)$ [Extremities of latus rectum of C_1]

Possible equation of direction of $C_2: y-a=a\Rightarrow y-2a=0$

$y+a+a=0\Rightarrow y+2a=0$

42. $A+B=ABAB=A^2+A$

$\Rightarrow B=A^2$

$BAB=A^5=A+I\Rightarrow A\left(A^4-I\right)=I$

$B^5-A^5=\left(A^5\right)^2-A^5=(A+I)^2-(A+I)=A^2+A=A+B$

43. $\vec{a}.b+b.\vec{c}=\vec{a}.\vec{c}+b.\vec{c}=\vec{a}.b+\vec{c}.b=0$

$\Rightarrow \vec{a},\vec{b},\vec{c}$ are mutually $\perp^{ar}$.

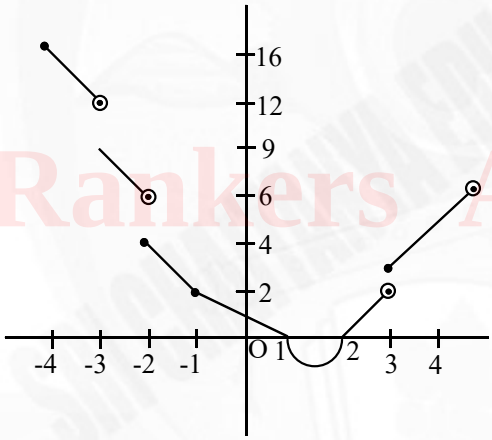
$\Rightarrow \vec{a}=x\hat{i},\vec{b}=y\hat{j},c=(x+y)\hat{k}$

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$$\frac{\left|4\vec{a}+4\vec{b}+3\vec{c}\right|^2}{--}=\frac{25\left(x^2+y^2\right)+18xy}{}\in[17,25] \text{ where } |\vec{a}|=x \,\&\, |\vec{b}|=y$$

44. $$f(x)=\begin{cases} x[x] & -4\le x<-1 \\ 1-x & -1\le x<1 \\ x^2-3x+2 & 1\le x<2 \\ (x-2)[x] & 2\le x\le 4 \end{cases}$$

$$f(x)=\begin{cases} -4x & x\in[-4,-3) \\ -3x & x\in[-3,-2) \\ -2x & x\in[-2,-1) \\ 1-x & x\in[-1,1) \\ x^2-3x+2 & x\in[1,2) \\ 2(x-2) & x\in[2,3) \\ 3(x-2) & x\in[3,4) \\ 8 & x=4 \end{cases}$$



1) $f(x)$ is DC at $x=-3,2,3,4\to 4$ points.

2) $f(x)$ is ND at $x=-3,-2,3,4,-1,2\to 6$ points.

3) Minimum at $x=\frac{3}{2}$, maxima at $x=-4,4$.

4) No. of solution of $f(x)=4$ is 2.

5) Sum of global maximum value and global minimum value of $\frac{63}{4}$.

45. $$\frac{x}{4x^2-2x+9}=\frac{1}{4x+\frac{9}{x}-2}\le\frac{1}{10}\left\{\because 4x+\frac{9}{x}\ge 12 \text{ here } x\in\left[\frac{1}{5},5\right]\right\}$$

$$I=\underbrace{\int_{1/5}^{5}\frac{1}{x}\left(\frac{x}{4x^2-2x+9}\right)dx}_{I_1}+\underbrace{\int_{1/5}^{5}\frac{1}{x}\left(\frac{x}{9x^2-2x+4}\right)dx}_{I2}+$$

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$$I_2=\int_{1/5}^{5}\frac{1}{x}\left\{-\frac{x}{9x^2-2x+4}\right\}dx$$

Putting $x=\frac{1}{t}\Rightarrow dx=-\frac{1}{t^2}dt$

$$I_2=\int_{5}^{1/5}t\left\{-\frac{\frac{1}{t}}{\frac{9}{t^2}-\frac{2}{t}+4}\right\}\frac{1}{t^2}dt=\int_{1/5}^{5}\frac{1}{t}\left\{-\frac{t}{4t^2-2t+9}\right\}dt$$

Now, $I=\int_{1/5}^{5}\frac{1}{x}\left(\left\{\frac{x}{4x^2-2x+9}\right\}+\left\{-\frac{x}{4x^2-2x+9}\right\}\right)dx$

$=I=\int_{1/5}^{5}\frac{1}{x}dx \qquad \{\because \{x\}+\{-x\}=1 \text{ when } x\in I\}$

$\therefore I=(\ln x)_{1/5}^{5}=2l\,n\,5=p\,lnq+3\tan^{-1}r$

$\Rightarrow p=2,q=5\ \&\ r=0$

$\therefore p+q+r=7$

46.

47. Let $f(\theta)=\cos 5\theta-\sin\theta$

$f(\theta)=0\Rightarrow\cos 5\theta=\sin\theta=\cos\left(\frac{\pi}{2}-\theta\right)$

$\Rightarrow 5\theta=2n\pi\pm\left(\frac{\pi}{2}-\theta\right) \qquad \ldots\ldots(1)$

Let $g(\theta)=\cot 4\theta-\tan 2\theta$

$g(\theta)=0\Rightarrow\cot 4\theta=\tan 2\theta=\cot\left(\frac{\pi}{2}-2\theta\right)$

$\Rightarrow\theta=\left(\frac{2p+1}{12}\right)\pi \qquad \ldots\ldots(2)$

$\forall\theta\in[0,\pi]$, common solutions of (1) and (2) are $\frac{\pi}{12}$ and $\frac{5\pi}{12}$

$\Rightarrow$ Total number of solutions is 4046

48. Operate $R_1\to R_1-\sec x,/r_3$.

$$f(x)=\begin{vmatrix}0 & 0 & \sec^2 x+\cot x.\cos ecx-\cos x\\ \cos^2 x & \cos^2 x & \cos ec^2 x\\ 1 & \cos^2 x & \cos^2 x\end{vmatrix}$$

After expanding and simplifying, we get

$f(x)=-\cos^5 x-\sin^2 x$

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Choice (1) is true and choice (2) is not correct.

Now $\int_0^{\pi/2} f(x)dx = \int_0^{\pi/2} -\cos^5 x - \int_0^{\pi/2} \sin^2 x dx = -\frac{8}{15} - \frac{\pi}{4}$

49. Obvious: $d = \sqrt{3^2 + 4^2 + 12^2} = 13$

50. $z = x + iy$

$|6x + 8y + 4| + |8x + 6y + 6| = 10 \Rightarrow 7x + 8y = 0,\ 7x + 8y = -10,\ x - y = 4,\ x - y = -6$

has $(-1,0)$ inside it & $(0,1)$ outside it.

51 & 52. (I)

$x = \frac{2009 \times 2008 \times 2007 + 1}{2008 \times 2007 + 2007} = 2008 + \frac{1}{2009 \times 2007}$

$[x] - 2008 = 0 \qquad (I) \to P$

(II)

$P(\sqrt{6}\cos\theta, \sqrt{2}\sin\theta)$

$6\cos^2\theta + 2\sin^2\theta = 2^2$

$4\cos^2\theta = 2$

$\cos^2\theta = \frac{1}{2},\ \theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

$[\theta] = 0 \quad 2 \quad 3 \quad 5$

$(II) \to P, R, S, U$

III) $\frac{1}{e^2} + \frac{1}{(e')^2} = 1 \Rightarrow e' = \frac{5}{4}$

$4e' = 5 \qquad (III) \to (U)$

IV) Locus of P is directrix $x + 1 = 0$

Hence distance = 3 $\qquad (IV) \to (S)$

53 & 54. Variance $= 114 \Rightarrow \lambda = 6$ Mean $(\bar{x}) = \frac{\sum f_i x_i}{\sum f_i} = 24$

Median $= \frac{\left(\frac{N}{2}\right)^{th} + \left(\frac{N}{2}+1\right)^{th} Obser}{2} = 24$

Mean deviation $(\bar{x}) = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i} = 10$ Mean deviation $M = \frac{\sum f_i |x_i - M|}{\sum f_i} = 10$

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